

Data File Handling

Binary Files

- Binary files are capable to store large files such as records, images, etc.
- These files have a specific pattern to read data in correct type.
- We can open some binary files in the normal text editor but we can't read the content present inside the file. That's because all the binary files will be encoded in the binary format, which can be understood only by a computer or a machine.
- These files don't have any delimiter. They are easier and faster than text files.
- These files are capable in storing complex structures like list, dictionary etc. Python provides specific functions to handle these data types and data files.



Operations on Binary Files

- Following operations can be performed on binary files stored on the file system through our python program.
- Inserting/Appending record in a binary file.
- Read records from a binary file.
- Search a record in a binary file.
- Update a record in a binary file

Steps Involved in File Handling

Binary File Handling consists of following steps:

- import module (pickle)
- Open the file.
- Process file i.e perform read or write operation.
- Close the file



To Open a Binary File

Open(): It is a function with two parameters.

- file_name: It accepts a file name with .dat extension
- access_mode: It specifies the mode to access file . The default mode is reading mode .

Syntax-

`<file_object>=open(<file_name>,<access_mode>)`

- File object: It is just like a variable or object. It is holding the reference of disk file.

Example:

```
f=open ("Myfile .dat", "rb")
```

To Close a Binary File

Close() function breaks the link of file-object and the file on the disk.

- After close(), no task can be performed on that file through the file-object.

Syntax -

File_object.close()

Example:

f.close()

File Modes of a Binary File

- **rb** To read the file which is already existing.
- **rb+** To Read and write but the file pointer will be at the beginning of the file.
- **wb** Only writing mode, if file is existing the old file will be overwritten else the new file will be created.
- **wb+** To write and read, if file is existing the old file will be overwritten else the new file will be created
- **ab** Append mode, the file pointer will be at the end of the file.
- **ab+** Appending and reading if the file is existing then file pointer will be at the end of the file else new file will be created for reading and writing.

Pickling & Unpickling

- ❖ **Pickling** is the process of converting structure to a byte stream before writing to a file.
- ❖ While reading the content of file a reverse process called **Unpickling** is used to convert the byte stream back to the original format

❖ Creating a File

Function to write a record in the binary file (dump())

dump() : It is used to write the object/record in file which is loaded in binary mode

Syntax :

pickle.dump(object_to_write , file_object)

Example :

pickle.dump(record,f)

Example - dump()

Code File

```
Python 3.8.3 Shell
File Edit Format Run Options Window Help
#Creating binary file using pickle module
import pickle
record=[]
while True:
    rollno=int(input("Enter roll no:"))
    name=input("Enter name:")
    marks=int(input("Enter marks obtained"))
    data=[rollno,name,marks]
    record.append(data)
    ch=input("enter your choice(y/n)")
    if ch=="n"or ch=="N":
        break
F=open("d:/dfh/student.dat","wb")
pickle.dump(record,F)
print("Record Added")
F.close()
```

Output

```
>>>
=== RESTART: C:/Users/Akanksha/AppData/Local/Programs/Python/Python38-32/b1.py ===
Enter roll no:11
Enter name:Amit
Enter marks obtained95
enter your choice(y/n)y
Enter roll no:12
Enter name:Anika
Enter marks obtained98
enter your choice(y/n)n
Record Added
>>>
```

Binary File

```
data
28-08-2020 11:09 Text Document 1 KB
File Edit Format View Help
ËÏ•$ ]"["(KÏAmit"K_e]"(K•AAnika"Kbee.
```

Reading a File

Function to read a record from a binary File (load())

load() : it is used to read object from pickle file.

Syntax:

object = pickle.load(file_object)

Example:

s=pickle.load(f)

Example-load()

Code File

```
#Reading the binary file using pickle module

import pickle
F=open("d:/dfh/student.dat","rb")
std=pickle.load(F)
print("Content of Binary File")
for i in std:
    rno=i[0]
    name=i[1]
    marks=i[2]
    print(" Roll no=",rno)
    print("Name=",name)
    print("Marks=",marks)
F.close()
```

Example-load()

Output

```
Python 3.8.3 Shell
Python 3.8.3 (tags/v3.8.3:6f8c832, May 13 2020, 22:20:19) [MSC v.1925 32 bit (Intel)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
== RESTART: C:/Users/Akanksha/AppData/Local/Programs/Python/Python38-32/b2.py ==
Content of Binary File
Roll no= 11
Name= Amit
Marks= 95
Roll no= 12
Name= Anika
Marks= 98
>>> ]
```

Binary File

```
data
28-06-2020 11:09 Text Document 1 KB
student - Notepad
File Edit Format View Help
[{"(j'('K\\A\\Amit'K_e]')(K\\A\\Anika'Kbee.
```